

Data Handling & Regression – Acoustic Fire Extinguisher

This assignment uses the dataset as given in `Acoustic_Extinguisher_Fire_Dataset.xlsx`. This dataset is also available via [Acoustic Extinguisher Fire Dataset](#).

Perform the exercises given below in either MATLAB or Python. Upload in Portflow your code (in the formats `.m`, `.mlx`, `.py`, `.ipynb`) and a concise report stating your methods, results and conclusions (in `.pdf`-format).

These exercises are formulated more broadly than the exercises in the previous assignments. This is on purpose. It is up to you to make choices in the analysis. Not all methods from this course will be equally suited to be applied to this dataset.

Exercises

1. Read the online documentation for this dataset. Load and clean the data. Check whether the data adheres to the specifications given online.
2. Investigate how the variable `STATUS` depends on the controlled and uncontrolled parameters. Take into account the different fuel types. Explain whether regression would be suitable to model the relationship between parameters and outcome, and why. If yes, explain which type of regression.
3. Build, using feature selection, a regression model that explains the variable `STATUS` depending on the other variables. Explain your choices and how you got to the final model. Quantify the quality of the final model.
4. Using linear discriminant analysis, build another model. Do you see any clear differences with the previous model?
5. Using classification trees and/or classification forests, build a third model. Make the model easily explainable to others. Also for this model, quantify the quality.
6. Look back and evaluate: are there any aspects that could be improved regarding the collection of this dataset? If you would need to analyze the effectiveness of this fire extinguishing method, what would you have done differently? Is there any data that you are missing?